



Modelling Financial Markets

Seminar 12: Options and the Volatility Surface — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ ATM: At-The-Money
- ▣ CI: Confidence Interval
- ▣ CRR: Cox–Ross–Rubinstein (binomial tree)
- ▣ DM: Diebold–Mariano (test)
- ▣ DVOL: Deribit Volatility Index
- ▣ GBM: Gradient Boosting Machine
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ OTM: Out-of-The-Money
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index
- ▣ VIX3M: Cboe 3-Month Volatility Index
- ▣ VRP: Variance Risk Premium

Today's Question and Required Outputs

- Question of the seminar: **how much can we trust the numbers that option prices give us**: Greeks, implied volatility, a fitted smile and the variance risk premium?
- Route
 - ▶ A5, A7 on paper: one identity for all Greeks; implied volatility by Newton
 - ▶ B1: hedging error by simulation; B3: an SVI smile for Bitcoin; B5: the variance risk premium with Newey–West errors
 - ▶ C2: find the errors in an AI answer; B6 at home
- Required outputs (checklist on the last core slide): the A answers, three figures (B1, B3, B5) and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A5	why are gamma and vega of one expiry proportional?	Solved	–
A7	which volatility reproduces a market price of 4.50?	Solved	–
B1	how fast does the hedging error shrink with rebalancing?	Solved	–
B3	how precisely does SVI describe the Bitcoin smile?	Solved	–
B5	is the S&P 500 variance risk premium significant and stable?	Solved	–
C2	what is wrong in an AI answer?	Proposed	A5, A7
B6 (home)	is the VIX an efficient forecast of volatility?	Proposed	B5
Extension packs	A1–A4, A6, A8, B2, B4, B7, B8, C1, C3 (optional, after the seminar)	both	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Used

Series	Symbol, field	Period	Units	Used in
S&P 500	GSPC.INDX, close	1990–2026	index points	B2, B5, B6, C1, C3
VIX, VIX3M	VIX.INDX, VIX3M.INDX, close	1990 / 2007–2026	annualised %	B2, B5, B6, C1
Bitcoin	BTC-USD.CC, close	2021–2026	USD	B7, C1
Bitcoin options	Deribit option snapshot: strike, type, bid, ask, mark price (BTC), mark IV, forward, index	22 Sep 2026, 07:51 UTC	BTC; % for IV	A1, A2, B3, B4, B8
DVOL	Deribit volatility index, daily	2021–2026	annualised %	B7, C1
Simulated	GBM paths, seed 42	–	–	B1

- Daily prices of the course data, one series per asset (field used: close), up to 18 September 2026; the Bitcoin options are a snapshot of Deribit quotes (the table of the chapter's Quantlet)
- Two series together: join the prices on common days first, then compute returns; C3 also needs the Kenneth French daily risk-free rate and the replication package of the case study

Setup: Run and Verify

```
# notebook section "Data" defines
# load_close(name) and deribit_chain()
px = pd.concat([load_close('sp500'),
                load_close('vix')],
               axis=1, join='inner').dropna()
print(len(px), px.index[0], px.index[-1])
print(px.iloc[[0, -1]])
c = deribit_chain()
print(len(c), c['expiry'].nunique())
```

- Expected output: 9 245 common days, 1990-01-02 to 2026-09-18

date	S&P 500	VIX
1990-01-02	359.69	17.24
2026-09-18	7650.50	14.81

- Option snapshot: 986 contracts (calls and puts), 11 expiries
- If your counts differ, stop and check the symbols, the join and the end date

Notation Used in Every Exercise

- S price of the underlying, K strike, τ time to expiry in years, r continuous rate, σ volatility (a decimal in formulas, in % when quoted); F forward, $k = \ln(K/F)$ log-moneyness, $w = \sigma_{imp}^2 \tau$ total implied variance
- Black–Scholes [Black & Scholes \(1973\)](#): $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$; Φ , φ the distribution function and density of the standard Normal distribution
- Greeks: $\Delta = \partial C/\partial S$, $\Gamma = \partial^2 C/\partial S^2$, vega $\mathcal{V} = \partial C/\partial \sigma$ per unit of σ (per vol point: divide by 100); “vol points” = percentage points of volatility
- Realised variance $RV_{t,t+h} = \frac{A}{h} \sum_{i=1}^h r_{t+i}^2$ ($A = 252$ trading or 365 calendar days); variance risk premium $VRP_t = IV_t^2 - RV_{t,t+h}$, in vol points $IV_t - \sqrt{RV_{t,t+h}}$ [Carr & Wu \(2009\)](#)
- Overlapping h -day windows are autocorrelated: Newey–West standard errors with at least h lags [Newey & West \(1987\)](#), or a block bootstrap [Künsch \(1989\)](#)
- One worked line: $S = K = 100$, $\tau = 0.5$, $r = 3\%$, $\sigma = 25\%$: $\sigma\sqrt{\tau} = 0.1768$, $d_1 = 0.1732$, $d_2 = -0.0035$, $C = 7.76$

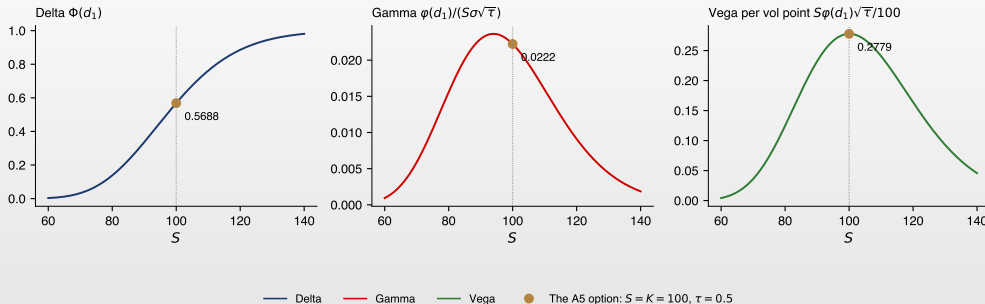
A5 [Solved]: One Identity for All the Greeks — Task

- **Question:** why are the gamma and the vega of options with the same expiry proportional, and what are they for an at-the-money call?
- **Inputs**
 - ▶ The Black–Scholes call of the notation slide; no dividends
 - ▶ Numbers: $S = K = 100$, six months ($\tau = 0.5$), $r = 3\%$, $\sigma = 25\%$; no data file
- **Tasks**
 1. Prove the identity $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ by computing $d_1^2 - d_2^2$.
 2. Use the identity to show that $\Delta = \Phi(d_1)$ and vega $= S\varphi(d_1)\sqrt{\tau}$.
 3. Show that vega $= \Gamma S^2\sigma\tau$, with $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$.
 4. Compute d_1 , d_2 , both sides of the identity, Δ , Γ , vega in two ways, the call, and the put from put–call parity.
- **Report:** the three proofs, a table of the numbers, and one sentence on what vega $= \Gamma S^2\sigma\tau$ implies for hedging gamma with options of the same expiry
- **Notebook:** section A5

A5 [Solved]: Solution (1/2) — Proofs

- Step 1: $\varphi(d_2)/\varphi(d_1) = e^{(d_1^2 - d_2^2)/2}$ and $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{\tau} \cdot \frac{2\ln(S/K) + 2r\tau}{\sigma\sqrt{\tau}}$
- Hence $\varphi(d_2)/\varphi(d_1) = Se^{r\tau}/K$, i.e. $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- Step 2: $\partial_S C = \Phi(d_1) + [S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)]\partial_S d_1 = \Phi(d_1)$, since $\partial_S d_1 = \partial_S d_2$ and the bracket is 0 by Step 1
- Vega: $S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
- Step 3: $\Gamma S^2\sigma\tau = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} S^2\sigma\tau = S\varphi(d_1)\sqrt{\tau} = \text{vega}$

A5 [Solved]: Solution (2/2) — Numbers and Chart



- Step 4: $\sigma\sqrt{\tau} = 0.1768$, $d_1 = (0.03 + 0.03125) \cdot 0.5 / 0.1768 = 0.1732$, $d_2 = -0.0035$; identity: $100 \cdot 0.3930 = 39.30$ and $98.51 \cdot 0.3989 = 39.30$
- $\Delta = 0.5688$, $\Gamma = 0.0222$; vega = $100 \cdot 0.3930\sqrt{0.5} = 27.79 = \Gamma S^2 \sigma \tau = 27.79$ per unit of σ (0.2779 per vol point); $C = 7.76$, $P = C - 100 + 98.51 = 6.27$
- **Interpretation:** for one expiry vega and gamma are proportional, so a gamma-neutral book of same-expiry options is also vega-neutral, but only while the smile is flat (extension A6)

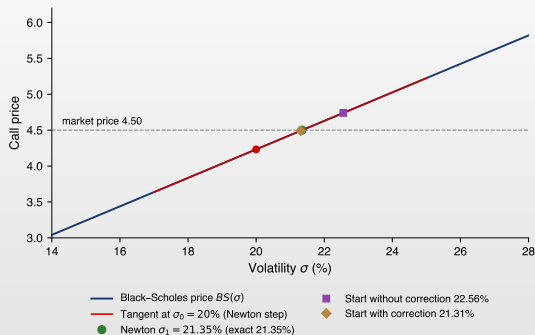
A7 [Solved]: Implied Volatility by Newton's Method — Task

- **Question:** which volatility makes the Black–Scholes price equal to a market price of 4.50, and how fast does Newton's method find it?
- **Inputs:** a European call with $S = K = 100$, three months ($\tau = 0.25$), $r = 2\%$, market price 4.50; vega $\mathcal{V}(\sigma) = S\varphi(d_1)\sqrt{\tau}$ from A5
- **Tasks**
 1. Prove that $\sigma \mapsto BS(\sigma)$ is continuous and strictly increasing, with limits $(S - Ke^{-r\tau})^+$ as $\sigma \rightarrow 0$ and S as $\sigma \rightarrow \infty$, and state when an implied volatility exists.
 2. Derive the at-the-money-forward approximation $C \approx S\sigma\sqrt{\tau}/\sqrt{2\pi}$ [Brenner & Subrahmanyam \(1988\)](#).
 3. Invert it to get two starting values: from the price 4.50, and from 4.50 minus half the discounted forward intrinsic value $(S - Ke^{-r\tau})/2$.
 4. Run one Newton step $\sigma_1 = \sigma_0 - (BS(\sigma_0) - 4.50)/\mathcal{V}(\sigma_0)$ from $\sigma_0 = 20\%$ and compare σ_1 with the exact implied volatility.
- **Report:** the two starting values, σ_1 and the exact implied volatility; one sentence on why Newton converges so fast here and when it would fail
- **Notebook:** section A7

A7 [Solved]: Solution (1/2) — Existence and Approximation

- Step 1: $\partial_\sigma BS = S\varphi(d_1)\sqrt{\tau} > 0$ (A5): strictly increasing and continuous
- $\sigma \rightarrow 0$: $d_{1,2} \rightarrow \pm\infty$ according to the sign of $\ln(S/K) + r\tau$, so $BS \rightarrow (S - Ke^{-r\tau})^+$; $\sigma \rightarrow \infty$: $d_1 \rightarrow \infty$, $d_2 \rightarrow -\infty$, so $BS \rightarrow S$
- By the intermediate value theorem a unique σ_{imp} exists if and only if the price lies strictly inside the no-arbitrage bounds $(S - Ke^{-r\tau})^+ < C < S$
- Step 2: if $S = Ke^{-r\tau}$, $d_{1,2} = \pm\sigma\sqrt{\tau}/2$ and $C = S[\Phi(\sigma\sqrt{\tau}/2) - \Phi(-\sigma\sqrt{\tau}/2)] \approx S\sigma\sqrt{\tau}\varphi(0) = S\sigma\sqrt{\tau}/\sqrt{2\pi}$

A7 [Solved]: Solution (2/2) — Newton Step and Chart

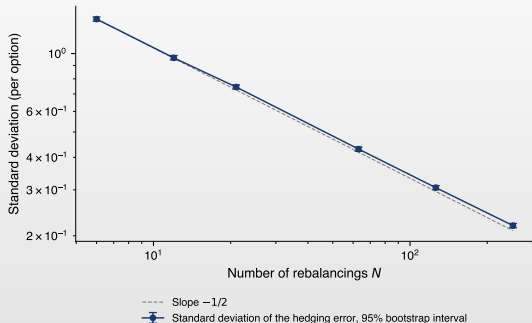


- Step 3: without correction
 $\sigma \approx 4.50\sqrt{2\pi}/(100 \cdot 0.5) = 22.56\%$
- With correction: $Ke^{-r\tau} = 99.501$, half the forward intrinsic value 0.249,
 $\sigma \approx (4.50 - 0.249)\sqrt{2\pi}/50 = 21.31\%$
- Step 4: $BS(20\%) = 4.2322$, $\mathcal{V} = 19.848$, so
 $\sigma_1 = 0.20 - (4.2322 - 4.50)/19.848 = 21.3495\%$;
 exact 21.3495%
- **Interpretation:** near the money the price is almost linear in σ , so one step gives four decimals; deep out of the money vega ≈ 0 and Newton can overshoot: use a bracketing solver (Brent)

B1 [Solved]: Hedging Error and Rebalancing Frequency — Task

- **Question:** how fast does the hedging error of a delta-hedged call shrink as the dealer rebalances more often?
- **Inputs**
 - ▶ Simulated data: 10 000 paths of geometric Brownian motion, $S_0 = 100$, $\mu = 8\%$, $\sigma = 20\%$, 252 steps over three months (seed 42)
 - ▶ A three-month ATM call ($K = 100$, $r = 4\%$) sold at its Black–Scholes premium 4.49 and delta-hedged at $\sigma = 20\%$
- **Tasks**
 1. For each $N \in \{6, 12, 21, 63, 126, 252\}$, rebalance the hedge N times and compute on every path the hedging error at expiry: shares plus cash minus the pay-off.
 2. Compute the standard deviation of the error for each N and a 95% percentile bootstrap interval from 1 000 resamples of the paths [Efron \(1979\)](#).
 3. Regress $\ln(\text{s.d.})$ on $\ln N$ and give the slope with a bootstrap interval from the same resamples.
 4. Find the smallest N in the list for which the standard deviation is below 10% of the premium.
- **Report:** a table of standard deviations with intervals, the log-log chart and the slope; one sentence on whether the slope is consistent with $-1/2$ and what transaction costs would change
- **Notebook:** section B1

B1 [Solved]: Solution (1/2) — Chart and Table



- Step 1: on each path, sell the call at 4.49, buy Δ_0 shares, finance with cash at r ; at each rebalancing date reset Δ ; at expiry, error = shares + cash - pay-off
- Step 2: the s.d. falls from 1.356 ($N = 6$) to 0.219 ($N = 252$); the intervals are narrow because 10 000 paths give a precise s.d.

N	6	12	21	63	126	252
s.d.	1.356	0.963	0.744	0.430	0.306	0.219
95% CI	[1.331, 1.381]	[0.945, 0.983]	[0.731, 0.758]	[0.422, 0.438]	[0.300, 0.312]	[0.215, 0.223]

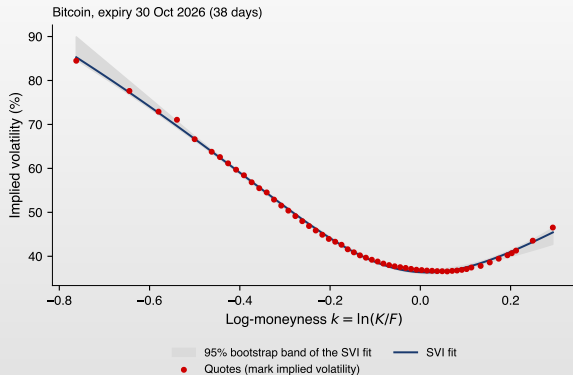
B1 [Solved]: Solution (2/2) — Slope and Interpretation

- Step 3: slope -0.489 , 95% CI $[-0.496, -0.482]$: close to $-1/2$ and slightly above it at these finite N ; the $1/\sqrt{N}$ law of the lecture is asymptotic
- Step 4: daily hedging ($N = 63$) leaves 9.6% of the premium, about the 10% target; four times a day halves it again (ratio 1.97)
- **Interpretation:** with a fixed fee per trade, costs grow like N while the error falls like $1/\sqrt{N}$; the best frequency depends on the cost model and on the trade-off chosen between risk and cost

B3 [Solved]: An SVI Smile for Bitcoin — Task

- **Question:** does a five-parameter SVI curve describe the Bitcoin smile, and how precisely are its at-the-money level and its skew estimated?
- **Inputs**
 - ▶ Option snapshot of the lecture; expiry closest to 30 days: 2026-10-30, 38 days, 56 quoted OTM options (puts below the forward, calls above); for each quote $k = \ln(K/F)$ and $w = \sigma_{\text{mark}}^2 \tau$, with σ_{mark} the mark IV
 - ▶ SVI Gatheral & Jacquier (2014): $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + s^2})$; the butterfly function $g(k)$ of the lecture: $g \geq 0$ for all k means a nonnegative density
- **Tasks**
 1. Fit the five SVI parameters by least squares on w , with $b \geq 0$, $|\rho| < 1$, $s > 0$, and report the root mean squared error in vol points.
 2. Resample the quotes with replacement 300 times, refit SVI each time, and report 95% intervals for the ATM volatility and the skew $IV(0.15) - IV(-0.15)$.
 3. Evaluate $g(k)$ on $k \in [-1.5, 1.5]$ and report its minimum.
- **Report:** parameters, intervals, min g and the chart of quotes, fit and band; one sentence each: which parameter controls the skew and its sign; is the skew significantly different from 0; why fit w and not prices
- **Notebook:** section B3

B3 [Solved]: Solution (1/2) — Fit and Chart



- Step 1: $\hat{a} = -0.020$, $\hat{b} = 0.102$, $\hat{\rho} = -0.262$, $\hat{m} = -0.079$, $\hat{s} = 0.348$; root mean squared error 0.42 vol points
- Step 2: ATM volatility 36.37%, 95% CI [36.10, 36.87]; skew -2.11 , 95% CI $[-2.73, -1.52]$
- Step 3: $\min_k g(k) = 0.136 > 0$ on $k \in [-1.5, 1.5]$: no butterfly arbitrage on this grid

B3 [Solved]: Solution (2/2) — Interpretation

- ▣ ρ controls the tilt: $\hat{\rho} < 0$ makes the put wing dearer; the skew interval excludes 0, so the tilt is significant at 5%
- ▣ Why w : it is comparable across strikes and nearly linear in the wings, while option prices span four orders of magnitude and would weight the fit towards the dearest options
- ▣ A positive density everywhere also needs $g \geq 0$ outside the grid and the wing conditions of Gatheral & Jacquier (2014)
- ▣ **Interpretation:** the bootstrap resamples smoothed mark volatilities, so the intervals measure fitting noise, not bid–ask uncertainty: they are a lower bound on the true uncertainty

B5 [Solved]: The Variance Risk Premium of the S&P 500 — Task

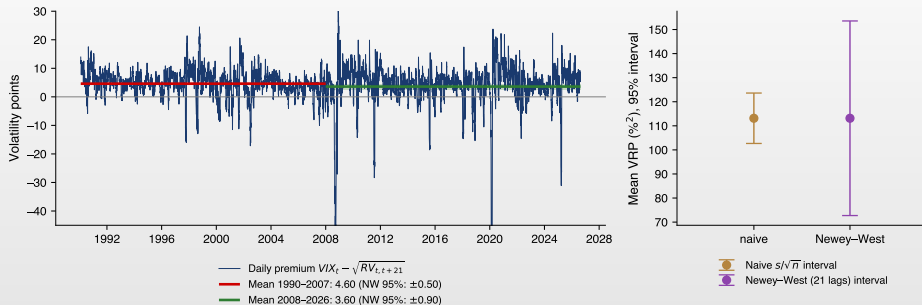
- **Question:** is the variance risk premium of the S&P 500 significantly positive, and was it the same before and after 2008?
- **Inputs**
 - ▶ Daily S&P 500 and VIX closes joined on common days, January 1990 – August 2026 (the last 21-day window ends in September 2026)
 - ▶ $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ in $\%^2$; $VRP_t = VIX_t^2 - RV_{t,t+21}$ in vol points $VIX_t - \sqrt{RV_{t,t+21}}$ Carr & Wu (2009)
- **Tasks**
 1. Build $RV_{t,t+21}$ from the next 21 log returns and VRP_t on every day with a complete window.
 2. Compute the mean VRP, its Newey–West standard error with 21 lags Newey & West (1987), a 95% interval, and the naive standard error $s/\sqrt{n}$.
 3. Repeat step 2 in vol points and compute the share of days with a positive premium.
 4. Repeat for 1990–2007 and 2008–2026, then test equality by regressing VRP_t on a constant and a 2008–2026 dummy with Newey–West errors (21 lags).
- **Report:** a table by period and the chart; one sentence each: why the naive standard error is wrong here and by how much; is the premium stable; why is the variance premium less precise than the vol-point premium?
- **Notebook:** section B5

B5 [Solved]: Solution (1/2) — Table and Steps

	n	Mean VRP (% ²)	95% CI	Vol points (s.e.)	Positive
1990–2026	9 203	113.1	[72.7, 153.6]	4.09 (0.27)	86%
1990–2007	4 516	149.3	[125.2, 173.5]	4.60 (0.25)	88%
2008–2026	4 687	78.3	[2.7, 153.9]	3.60 (0.46)	84%

- Step 1: $RV_{t,t+21}$ from the next 21 log returns, then VRP_t on every day with a complete window
- Step 2: regress VRP_t on a constant with HAC standard errors (Bartlett, 21 lags: the overlap); interval = mean ± 1.96 s.e.; full sample $t = 113.1/20.6 = 5.49$: reject H_0 at 5%
- Naive s.e. 5.3 vs Newey–West 20.6: 3.9 times too small, because consecutive windows share 20 of 21 days

B5 [Solved]: Solution (2/2) — Chart and Interpretation



- Step 3: vol points 4.09 (s.e. 0.27), positive on 86% of days; Step 4: 1990–2007 $t = 12.12$, 2008–2026 $t = 2.03$; difference $-71.0\% ^2$ ($p = 0.08$), in vol points -1.01 ($p = 0.06$)
- Interpretation:** squaring magnifies the crisis windows (2008, 2020), so the variance premium has heavier tails and a wider interval than the vol-point premium; the premium is positive in both subperiods, but the difference in size is not significant at 5%: the data cannot tell whether it changed

C2 [Proposed]: Critique an AI Answer — the Answer

- A student asks an AI assistant: “Price a European call with 90 calendar days to expiry on a non-dividend-paying stock with $S = 100$, $K = 105$, volatility 20%, continuous rate $r = 4\%$; then give the put from put–call parity”
- The assistant answers with this code:

```
import numpy as np
from scipy.stats import norm
S, K, r, sigma = 100.0, 105.0, 0.04, 20          # sigma = 20%
T = 90 / 252                                     # 90 days
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put = call + S - K * np.exp(-r * T)             # put-call parity
print(f"call_{S}={call:.2f},_{K}put_{S}={put:.2f}")
```

- “The code follows Black–Scholes exactly; the put comes from parity, so no second formula is needed”

C2 [Proposed]: Critique an AI Answer — Task

Solution discussed in class

- **Question:** which three errors does the AI answer contain, and which check catches each of them?
- **Inputs:** the code on the previous slide; everything except three lines is correct
- **Tasks**
 1. Before running the code, estimate what a 90-day call 5% out of the money should cost: 0.2, 2 or 20.
 2. For each error, say what is wrong and why, and how you would detect it: a no-arbitrage bound, a unit check, the lecture, or a second method.
 3. Correct the three lines, run the code, report C and P , and check put–call parity numerically.
 4. Log the three errors in `AI_ERRORS.md` with the fields: what was asked, what the AI answered, what was wrong, how you found out, the correct version.
- **Report:** the corrected code and output; one sentence on which error still gives a plausible-looking price, and one on why “the put comes from parity” is not a check of the call
- **Model:** A5, A7 (Black–Scholes step by step)

B6 [Proposed]: Is the VIX an Efficient Forecast? — Task

Solution discussed in class

- **Question:** is VIX_t^2 an unbiased and efficient forecast of the variance realised over the next 21 trading days?
- **Inputs**
 - ▶ The daily series of B5: $RV_{t,t+21}$ and VIX_t^2 , 1990–2026; past variance $RV_{t-21,t}$
 - ▶ For step 4, a log HAR-RV [Corsi \(2009\)](#): $\ln RV_{t,t+21}$ on the logs of the 1, 5 and 22-day averages of daily squared returns (annualised, %², each floored at 0.01); level forecast $\exp(\widehat{\ln RV} + \hat{s}^2/2)$
- **Tasks**
 1. Estimate $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ [Mincer & Zarnowitz \(1969\)](#) with Newey–West errors (21 lags); test $\alpha = 0$ and $\beta = 1$ separately and jointly (Wald, 5%), and report $\text{corr}(\hat{\alpha}, \hat{\beta})$.
 2. Add $RV_{t-21,t}$ as a second regressor; then re-estimate in logs, $\ln RV$ on $\ln VIX^2$, and test $\beta = 1$.
 3. Instrument $\ln VIX_t^2$ with $\ln VIX_{t-21}^2$ [Christensen & Prabhala \(1998\)](#) (just-identified GMM, HAC covariance); report the first-stage t and test $\beta = 1$.
 4. From 2000, re-estimate every 21 days on an expanding window of observed targets; compare VIX^2 with HAR-RV by QLIKE loss and a Diebold–Mariano test with HAC (21 lags) [Diebold & Mariano \(1995\)](#), [Patton \(2011\)](#).
- **Report:** all estimates and tests with decisions at 5%; one sentence each: why both t -statistics can be small while the joint test rejects; what the instrument must satisfy; does in-sample significance survive out of sample?
- **Model:** B5 (Newey–West inference)
- **Notebook:** section B6

Submission Checklist

- ☐ Hand in (team repository, folder seminar12/)
 - ▶ A5, A7: the proofs and the tables of numbers (photo or $\text{\LaTeX}$)
 - ▶ B1, B3, B5: one figure and one table each, with the interpretation sentence
 - ▶ C2: the corrected code, its output and the three entries in `AI_ERRORS.md`; B6: all tests with decisions at 5%
- ☐ The executed notebook; every prompt that shaped a decision in `AI_USE.md`
- ☐ Check before submitting
 - ▶ Dividends and the interest rate are in parity; vega and theta carry their units (per vol point, per day)
 - ▶ No naive standard errors on overlapping windows; no risk-neutral probability read as a forecast
 - ▶ A Sharpe ratio of a short-volatility strategy is reported with its skewness and worst month
- ☐ Team-project seed: repeat B3, B4 and B8 on a new Deribit snapshot of Ether options and compare with Bitcoin [Pele et al. \(2023\)](#); delta-hedge one-week Bitcoin options on historical paths [Matic, Packham & Härdle \(2023\)](#)

Extension Packs: What Each One Needs

Exercise	Question	Type	Model	Data
A1	are Bitcoin call prices convex in the strike?	Solved	–	option snapshot
A2	which forward and rate does parity imply?	Proposed	A1	option snapshot
A3	what does a dealer earn when $\sigma_{real} \neq \sigma_{imp}$?	Solved	–	–
A4	how fast does the binomial tree converge?	Proposed	A3	–
A6	can one option remove gamma and vega together?	Proposed	A5	–
A8	how large are the errors of a variance strip?	Proposed	A1	–
B2	did delta-hedged option sellers earn a premium?	Proposed	B1, B5	S&P 500, VIX
B4	what is a VIX for Bitcoin?	Proposed	A8, B3	option snapshot
B7	is there a variance risk premium in Bitcoin?	Proposed	B5	DVOL, Bitcoin
B8	how fat are the tails the market prices?	Proposed	B3	option snapshot
C1	is the variance premium a tradable signal?	Proposed	B5, B7	S&P 500, VIX, DVOL
C3	is the Martin bound valid and tight?	Proposed	B5	case-study files

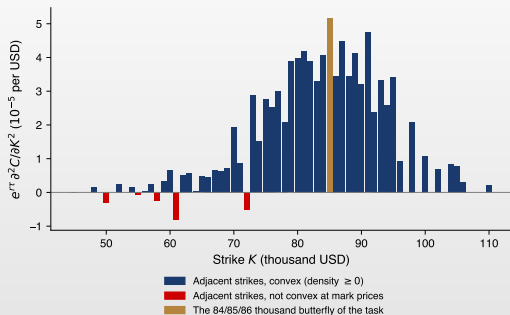
A1 [Solved]: Bounds and Convexity in the Strike — Task

- **Question:** which model-free restrictions must call prices satisfy across strikes, and do Bitcoin quotes satisfy them?
- **Inputs**
 - ▶ European calls on a non-dividend-paying asset, price $C(K)$, maturity τ , rate r ; no arbitrage, no frictions
 - ▶ Bitcoin calls expiring 30 October 2026 in the option snapshot; USD price = BTC premium $\times$ index $S = 85\,265$; forward $F = 85\,742$, so $e^{r\tau} = F/S$; **mark** = the exchange's reference price, **bid/ask** = the best buy and sell quotes
- **Tasks**
 1. Prove $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ by comparing the pay-offs of two portfolios in every state at T .
 2. For $K_1 < K_2 < K_3$ with $K_2 = \lambda K_1 + (1 - \lambda)K_3$, prove $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$.
 3. Show that the butterfly $[C(K - h) - 2C(K) + C(K + h)]/h^2$ tends to $e^{-r\tau}q(K)$, with q the risk-neutral density [Breedon & Litzenberger \(1978\)](#).
 4. Compute the 84/85/86 thousand butterfly from mark prices, $q(85\,000)$ and $P^Q(84\,000 < S_T < 86\,000)$; then count the negative butterflies among all 732 equally spaced triples, at mark prices and at executable prices (wings at the ask, body at the bid).
- **Report:** the proofs, the butterfly, $q(85\,000)$ and the violation counts; is any violation an arbitrage you could trade?
- **Notebook:** section A1

A1 [Solved]: Solution (1/2) — Proofs

- Step 1: a call plus $Ke^{-r\tau}$ in cash is worth $\max(S_T, K) \geq S_T$ at T , so $C + Ke^{-r\tau} \geq S$; the call pays $(S_T - K)^+ \geq 0$, so $C \geq 0$
 - ▶ Upper bound: the share pays $S_T \geq (S_T - K)^+$ in every state, so $C \leq S$
- Step 2: for fixed S_T the map $K \mapsto (S_T - K)^+$ is convex, so $\lambda(S_T - K_1)^+ + (1 - \lambda)(S_T - K_3)^+ \geq (S_T - K_2)^+$ in every state
 - ▶ A portfolio that pays at least as much in every state costs at least as much today
- Step 3: $C(K) = e^{-r\tau} \int_K^\infty (x - K)q(x) dx$, so $\partial_K C = -e^{-r\tau} \int_K^\infty q(x) dx$ and $\partial_K^2 C = e^{-r\tau} q(K)$
 - ▶ The butterfly quotient is the central second difference, so it converges to $e^{-r\tau} q(K)$: convexity is exactly $q \geq 0$

A1 [Solved]: Solution (2/2) — Bitcoin Numbers and Chart



- Step 4: $C(84\,000) = 4957.26$, $C(85\,000) = 4414.38$, $C(86\,000) = 3922.98$ USD; butterfly = 51.47 USD > 0
- $q(85\,000) \approx 1.0056 \cdot 51.47 / 1\,000^2 = 5.18 \cdot 10^{-5}$ per USD; $P^Q(84\,000 < S_T < 86\,000) \approx 2h q(K) = 10.4\%$
- 7 of 732 mark butterflies are negative (worst -11.80 USD, deep in the money, median spread 255.80 USD); at executable prices: 0
- Interpretation:** no tradable arbitrage, but a density read from raw marks is negative in places (red bars, 5 of 54 adjacent triples): smooth the smile before differentiating

A2 [Proposed]: Parity as a Regression: the Implied Forward — Task

Solution discussed in class

- ▣ **Question:** which forward price and which USD interest rate do Bitcoin option prices imply through put–call parity?
- ▣ **Inputs:** the expiry of A1; USD mid prices $(bid + ask)/2 \times S$; only strikes where both the call and the put have a bid and an ask; $D = e^{-r\tau}$ and F unknown
- ▣ **Tasks**
 1. Show from put–call parity that $C(K) - P(K) = DF - DK$ for every strike.
 2. Write the OLS regression $C_i - P_i = a + bK_i + e_i$ and derive $\hat{D}$ and $\hat{F}$ from $(\hat{a}, \hat{b})$.
 3. Derive the delta-method standard error of $\hat{F}$ with a heteroskedasticity-robust covariance.
 4. Estimate on the data and compare $\hat{F}$ with the exchange forward, in standard errors.
- ▣ **Report:** $\hat{D}$, $\hat{F}$ with standard errors and the number of strikes; which USD rate does $\hat{D}$ imply, and why might $\hat{F}$ differ from the exchange forward?
- ▣ **Model:** A1 (portfolios with identical pay-offs)
- ▣ **Notebook:** section A2

A3 [Solved]: The Black–Scholes Equation and the Hedging P&L — Task

- **Question:** what does a dealer who sells an option at σ_{imp} and delta-hedges it earn when the price moves with σ_{real} ?
- **Inputs**
 - ▶ $dS_t = \mu S_t dt + \sigma S_t dW_t$, constant r , continuous trading without costs; option value $V(S, t)$ twice differentiable [Black & Scholes \(1973\)](#)
 - ▶ Numbers: the call of A5, sold at $\sigma_{imp} = 25\%$, gamma $\Gamma = 0.0222$ (from A5); 252 trading days a year
- **Tasks**
 1. Replicate V with a self-financing portfolio of Δ shares and cash B at r ; apply Itô's lemma, choose Δ to remove the dW_t term, and derive the Black–Scholes equation from no arbitrage.
 2. For a dealer who sells at σ_{imp} and holds $\Delta = \partial_S V^{imp}$ while S moves with σ_{real} , derive the instantaneous P&L.
 3. Compute today's P&L rate per trading day if $\sigma_{real} = 30\%$, then annualise it with S and Γ frozen.
- **Report:** the equation, the P&L formula and the two numbers; who gains, and why does the P&L depend on the path of S ?
- Notebook: section A3

A3 [Solved]: Solution (1/2) — The Equation

- Step 1: Itô: $dV = (V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t$; the self-financing $\Pi = \Delta S + B$, started at $\Pi = V$, has $d\Pi = \Delta dS + rB dt$
- $d(V - \Pi) = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB) dt + (V_S - \Delta)(\mu S dt + \sigma S dW_t)$; with $\Delta = V_S$ both the risk and μ disappear
- A riskless position that costs 0 earns 0: $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$
- Step 2: the book $-V + \Delta S + (V - \Delta S)$ in cash changes by $-dV + \Delta dS + r(V - \Delta S) dt$ plus interest on the P&L; Itô with σ_{real} and the equation at σ_{imp} give
- $dP\&L = r P\&L dt + \frac{1}{2}\Gamma S^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$

A3 [Solved]: Solution (2/2) — Numbers

- Step 3: $\frac{1}{2} \cdot 0.02223 \cdot 100^2 \cdot (0.25^2 - 0.30^2)/252 = \frac{1}{2} \cdot 0.02223 \cdot 10^4 \cdot (-0.0275)/252 = -0.0121$ per option and trading day
- Annualised with S and Γ frozen: -3.06 , about 39% of the call price 7.76: a rate, not the realised P&L of this six-month option, since ΓS^2 changes along the path
- **Interpretation:** the seller loses when realised exceeds implied volatility; the loss accrues through Γ , largest near the strike and near expiry, so two paths with the same realised variance give different P&L

A4 [Proposed]: The Binomial Tree Converges to Black–Scholes — Task

Solution discussed in class

- ▣ **Question:** how fast does the Cox–Ross–Rubinstein price approach the Black–Scholes price, and why does the error oscillate?
- ▣ **Inputs:** CRR tree with N steps of length $\Delta t = T/N$ Cox, Ross & Rubinstein (1979); European call of the lecture:
 $S = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$
- ▣ **Tasks**
 1. Show that $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $q = (e^{r\Delta t} - d)/(u - d)$ match the risk-neutral mean exactly and the variance of the log step up to $O(\Delta t^2)$.
 2. Price the call with $N = 50, 51, 100, 101, 200, 201$ and compute $N(C_N - C_{BS})$ for each.
 3. Average the prices for $N = 200$ and $N = 201$ and compare the error with that of $N = 200$.
- ▣ **Report:** a table of C_N and $N(C_N - C_{BS})$; the order of the error, the reason for the even/odd oscillation, and why averaging helps
- ▣ **Model:** A3 (from the tree to the continuous-time equation)
- ▣ **Notebook:** section A4

A6 [Proposed]: A Delta-Gamma Hedge — Task

Solution discussed in class

- ▣ **Question:** how many options and shares make a short option book delta- and gamma-neutral, and is it then vega-neutral too?
- ▣ **Inputs**
 - ▶ A dealer has **sold** 1 000 calls of A5 ($S = K = 100$, six months, $r = 3\%$, $\sigma = 25\%$)
 - ▶ Hedging instruments: a call with $K = 110$, same expiry, Black–Scholes price 3.90, and the stock; Greeks from the Black–Scholes formulas, gamma per unit of S
- ▣ **Tasks**
 1. Compute the delta and the gamma of the dealer's position.
 2. Find the number n_2 of $K = 110$ calls that makes the gamma zero, then the number of shares that makes the delta zero.
 3. Compute the vega of the hedged book and explain the result.
- ▣ **Report:** a positions table (instrument, quantity, delta, gamma, vega); is the book protected against a change in volatility?
- ▣ **Model:** A5
- ▣ **Notebook:** section A6

A8 [Proposed]: The Variance-Swap Strike and Its Estimation Error — Task

Solution discussed in class

- ▣ **Question:** how large are the errors of a finite strip of options when it is used to price variance, in a case where the answer is known?
- ▣ **Inputs**
 - ▶ Spanning formula of the lecture: $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$
Bakshi, Kapadia & Madan (2003)
 - ▶ Log-normal benchmark: $F = 100$, $r = 0$, 90 days ($T = 90/365$), flat volatility 20%, so the exact variance is 0.04; OTM prices from Black-Scholes; **discretisation** error = Cboe sum minus the dense integral on $[80, 120]$; **truncation** error = dense integral on $[80, 120]$ minus 0.04
- ▣ **Tasks**
 1. Apply the formula with $f(S) = -2 \ln(S/F)$ and show $K_{var} = \frac{2e^{rT}}{T} \left[\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right]$ for continuous paths.
 2. Compute the Cboe sum with $\Delta K = 5$ on $[80, 120]$ Cboe (2025), the dense integral on $[80, 120]$ and the dense integral on $(0, \infty)$.
 3. Split the total error into discretisation and truncation.
- ▣ **Report:** the three variances and volatilities and the two errors; which dominates here, and what changes for a longer maturity?
- ▣ **Model:** A1 (portfolios of options as prices of pay-off shapes)
- ▣ **Notebook:** section A8

B2 [Proposed]: Delta-Hedged S&P 500 Options, 1990–2026 — Task

Solution discussed in class

- ▣ **Question:** did sellers of delta-hedged one-month S&P 500 calls earn a significant premium, and at what tail risk?
- ▣ **Inputs**
 - ▶ Daily S&P 500 and VIX closes, January 1990 – September 2026; interest and dividends ignored
 - ▶ Strategy of the lecture: every 21 trading days sell a one-month ATM call priced with the VIX and hedge it daily at the same volatility; P&L in % of the index [Bakshi & Kapadia \(2003\)](#)
- ▣ **Tasks**
 1. Compute the mean monthly P&L and its Newey–West standard error with 3 lags, and test H_0 : mean = 0 at 5%.
 2. Build a 95% moving-block bootstrap interval for the mean (blocks of 6 months, 2 000 resamples) [Künsch \(1989\)](#).
 3. Compute the share of positive months, the skewness, and the share of the total gain lost in the five worst months.
- ▣ **Report:** the mean with both intervals and the tail statistics; why can a strategy with a high t -statistic still be dangerous?
- ▣ **Model:** B1 (bootstrap), B5 (Newey–West)
- ▣ **Notebook:** section B2

B4 [Proposed]: A VIX for Bitcoin — Task

Solution discussed in class

- **Question:** what does the Cboe VIX formula give for Bitcoin, and why does it differ from the at-the-money volatilities and from DVOL?
- **Inputs:** the option snapshot; the two expiries around 30 days (2026-10-09, 2026-10-30); USD mid $Q = \text{BTC mid} \times \text{index}$; F and $e^{RT} = F/S$ from the Deribit forward
- **Tasks**
 1. Select the strip of each expiry with the Cboe rules [Cboe \(2025\)](#): K_0 the first strike at or below F ; puts below K_0 , calls above, the average of both at K_0 ; drop zero bids and stop each wing after two consecutive zero bids.
 2. Compute each implied variance $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} (F/K_0 - 1)^2$, with ΔK_i half the distance between neighbouring strikes.
 3. Interpolate the total variances linearly in time to 30 days and report $100\sigma_{30}$.
 4. Compare with the ATM volatilities of the two expiries and with DVOL at the same moment [Alexander & Imeraj \(2020\)](#).
- **Report:** the two variances, the weight of the near expiry and the index; why is the index above both ATM volatilities, and which choices could explain the gap to DVOL?
- **Model:** A8 (strip formula and its errors), B3 (the same snapshot)
- **Notebook:** section B4

B7 [Proposed]: The Variance Risk Premium of Bitcoin — Task

Solution discussed in class

- **Question:** is the Bitcoin variance risk premium significantly positive, and is it larger than the equity premium?
- **Inputs:** DVOL (Deribit), March 2021 – August 2026, and daily Bitcoin closes (BTC-USD.CC); realised volatility over the next 30 calendar days, $A = 365$
- **Tasks**
 1. Compute the mean premium $DVOL_t - \sqrt{RV_{t,t+30}}$ in vol points and its Newey–West standard error with 30 lags, and test it at 5%.
 2. Compute the share of positive days and the ratio of mean implied to mean realised volatility, and compare both with the S&P 500 of B5.
 3. Count the non-overlapping 30-day periods in the sample.
- **Report:** premium, test, ratios and effective sample size; in which unit is the Bitcoin premium larger, and how much can five years of data tell us?
- **Model:** B5
- **Notebook:** section B7

B8 [Proposed]: A Risk-Neutral Density with Confidence Bands — Task

Solution discussed in class

- ▣ **Question:** how much more probability than the log-normal density does the Bitcoin market put on large moves, and how precise is that estimate?
- ▣ **Inputs:** the option snapshot; expiry closest to 90 days: 2026-12-25, 94 days, 58 OTM quotes; SVI fitted as in B3; forward (undiscounted) prices
- ▣ **Tasks**
 1. Compute the density $q(K) = \partial^2 C / \partial K^2$ from the SVI call prices (Black-76 with F) on a fine strike grid [Breedon & Litzenberger \(1978\)](#).
 2. Compute $P^Q(S_T < 0.8F)$ and $P^Q(S_T > 1.2F)$ with 95% pairs-bootstrap intervals (300 refits) and the log-normal values at the ATM volatility.
 3. In each refit, recompute the log-normal value at that refit's ATM volatility and give a 95% interval for the difference SVI minus log-normal.
 4. Report where the quotes end in K/F and which part of each probability relies on extrapolated wings.
- ▣ **Report:** the two probabilities with intervals vs log-normal; is the difference significant, and what does the bootstrap band not measure?
- ▣ **Model:** B3 (SVI fit and pairs bootstrap)
- ▣ **Notebook:** section B8

C1 [Proposed]: Is the Variance Risk Premium a Tradable Signal? — Task

Solution discussed in class

- ▣ **Question:** would you sell variance on the S&P 500 or on Bitcoin every month, and when would you stop? Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- ▣ **Inputs**
 - ▶ A synthetic variance swap with vega notional 1: monthly $P\&L = (IV^2 - RV)/(2 IV)$ vol points
 - ▶ S&P 500 with the VIX (1990–2026, VIX3M from 2007); Bitcoin with DVOL (2021–2026)
- ▣ **Tasks**
 1. Compute the monthly P&L series and the Sharpe ratio with a block-bootstrap interval, the skewness, the worst month and the maximum drawdown.
 2. Test a timing rule: skip months that start in backwardation (VIX above VIX3M); is it better?
 3. Check how fragile the result is to the day on which the monthly grid starts.
 4. Compare with the delta-hedged option of the lecture and propose a position size.
- ▣ **Report:** a table and a chart of cumulative P&L; a verdict with its limitations
- ▣ **Model:** B5, B7; the predictive regressions of the lecture appendix

C3 [Proposed]: Testing the Martin Bound — Task

Solution discussed in class

- **Question:** is the lower bound on the equity premium of [Martin \(2017\)](#) valid and tight on the course data, as in the case study [Back, Crotty & Kazempour \(2022\)](#)?
- **Inputs**
 - ▶ The authors' daily Martin bound, annualised %, $h = 1, 3, 6, 12$ months, 1990–2020 (columns `lb_m_h` of `market_bounds.csv`, [replication package](#))
 - ▶ S&P 500 (`GSPC.INDX`); daily risk-free rate ([Kenneth French Data Library](#)); monthly Welch–Goyal predictors [Welch & Goyal \(2008\)](#) ([Amit Goyal's data page](#))
- **Tasks**
 1. Compute the daily excess return (price return minus the risk-free rate), compound it over 21, 63, 126 and 252 trading days and annualise with $12/h$, for formation dates 31 January 1990 – December 2020 ($N = 7789$ at 1 month, 7598 at 12 months).
 2. Build the nine positive conditioning variables of Table 3 (carried forward from month-ends) and form the 10 moments $s_{t,T}(1, z_t)$ with $s_{t,T} = R_{t,T}^e - b_{t,T}$ and their mean $\bar{\lambda}$.
 3. Estimate Σ by Hansen–Hodrick [Hansen & Hodrick \(1980\)](#) with $21h$ lags (Newey–West with $1.5 \times 21h$ lags if not positive semi-definite) and compute D_1 , D_2 and their p -values [Kodde & Palm \(1986\)](#).
 4. Regress $R_{t,T}^e$ on a constant and $b_{t,T}$ (Hansen–Hodrick errors) and test $\alpha = 0$, $\beta = 1$ jointly; repeat step 3 with the bound replaced by its time-series mean (Section 4.3).
- **Report:** a table of means, D_1 , D_2 and p -values next to Tables 1 and 3, and one sentence on why the regression test and the tightness test can disagree
- **Model:** B5 (overlapping windows and HAC errors)

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